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Effect of water flow on the mobility of the root-knot nematode *Meloidogyne incognita* in columns filled with glass beads, sand or andisol

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ABSTRACT

In the present study, the migration of nematodes was studied in columns filled with three materials of different textures and chemical properties. The role of soil pores that enable root-knot nematode (*Meloidogyne incognita*) second stage juveniles (J2) to escape rapid water flow in soil was demonstrated using columns filled with glass beads, sand or andisol that maintained a constant water flow. Under a constant flow flux of 36 cm h^{-1}, living J2, dead J2 or anion bromine tracer (Br^-) was injected in the middle of the column and then drainage water equivalent to two pore volumes (PV) was collected. The passive transport of the anion tracer in water flow could be explained by a convection dispersion equation. The dead J2 showed a pattern similar to that of Br^-. However, the living J2 resisted movement in the water flow and remained in the column even at the highest water flow rate of 93.3 cm h^{-1} in glass beads. The mobility of living J2 was affected by the filling materials; the number of J2 passing through the column was much lower in the andisol-filled column than in the other two columns but the total number of J2 in drainage water was 5% or less of the number injected for all columns. We suggest that J2 were affected not only by soil water flow but also by soil pore structure and have the ability of withstanding or avoiding movement in soil water flow.

1. Introduction

Plant-parasitic nematodes parasitize roots and/or stems of various host plants affecting plant growth and development (Wyss and Zunke, 1998; Wyss et al., 1992). They often injure crops and the degree of damage due to nematode-infection was estimated to reach a hundred million dollars per year (Barker and Koenning, 1998). Among them the root-knot nematode (*Meloidogyne incognita*), which has a wide host range of over 700 species and varieties (Goodey et al., 1965), causes some of the most serious damage of over 10 billion dollars per year around the world, comprising more than 40% of total loss from plant-parasitic nematode-infections (Sasser and Freckman, 1987).

Plant-parasitic nematodes migrate through soil and they are subjected to many environments that influence their migration, development and survival. These include soil moisture (Wallace, 1968), which in turn, is affected by soil texture and structure affecting their biomass, survival, and migration (Wallace, 1958a,b; Hassink et al., 1993; Quénéhervé and Chotte, 1996). Factors including temperature (Griffin and Jensen, 1997), soil water content (Towson and Apt, 1983), oxygen concentration (Van Gundy and Stolzy, 1963), carbon dioxide concentration (Robinson, 1995), pH (Melakeberhan et al., 2004), soil structure (Eo et al., 2007), geotaxis (Eo et al., 2008), attractants (Castro et al., 1989; Hewlett et al., 1997), and repellants (Dies and Dusenbery, 1989; Castro et al., 1991) have been intensively studied.

The infective stage of *M. incognita* is the second stage juvenile (J2). Ranges of body length and width of the J2 are 360–393 μm and 10.9–13.6 μm, respectively, according to the original species description (Chitwood, 1949). The J2 are reported to migrate gradually into soil aggregates (Sano and Nakasono, 1997a,b) and the ability to migrate in soil depends upon the pore distribution (Otobe et al., 2004). All these suggest that soil conditions should be highly relevant to mobility of *M. incognita*, and thus to the infection efficiency. In the field, soil conditions are affected by water flows due to rainfall and irrigation and this also affects the ethology of nematodes as soil habitants. However, little is yet known about the relationship between nematode movement and soil water flow, although it is of importance in properly simulating nematode movement in soil water flows.

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The objective of the present study was to elucidate the influence of water flows in glass beads, sand or andisol with different pore distributions and structure on nematode movement and to compare nematode movement with solute transport in these water flows. For this purpose, we developed an apparatus that creates a uniform water flow in a column packed with glass beads, sand or soil into which nematodes were injected.

2. Materials and methods

2.1. Test materials

The test materials used were glass beads (diam. 800 μm), sand, and andisol (developed on volcanic ash; common soil in Japan). These materials differing in pore size distribution and presence of aggregates were used to compare the movement of *M. incognita* second stage juvenile (J2) in water flows. Stones, silt and clay present in the sand were removed, and then it was sieved to obtain a particle diameter size of 200–400 μm. Plant roots and stones were removed from the andisol immediately after collection from the field and it was handled carefully to prevent changes in moisture and soil characteristics. It was then sieved to obtain an aggregate size with a diameter of approximately 2000 μm. Glass beads and sand consist of single grains with a non-aggregated structure, while andisol has an aggregated structure. The composition of aggregates in the test andisol was analyzed as described by Yoder (1936) except for the diameter of sieves employed (2000 μm, 1500 μm, 500 μm, 250 μm, and 100 μm mesh). The different fractions were classified according to size: 1000–2000 μm (A2000); 500–1000 μm (A1000); 250–500 μm (A500); 100–250 μm (A250); 0–100 μm (A100).

All materials were examined for their moisture release properties in order to calculate the moisture release characteristic curve and the data obtained were used to estimate pore distribution by the following Eq. (1), which relates soil water matric potential to the neck diameter of the largest water filled pore (Marshall and Holmes, 1979).

$$\Psi_m = \frac{300}{d}, \tag{1}$$

where Ψ_m is the matric potential (kPa), and d is the diameter (μm) of the neck of the largest water filled pore. Pore size distribution per unit volume of glass beads, sand or andisol used in the experiment is shown in Fig. 1. The pores were classified into three pore size distributions of ≤20 μm, 20–130 μm, and ≥130 μm, since past studies indicated that root-knot nematodes (*Meloidogyne* spp.) move within pore diameters of approximately 20–130 μm as calculated from data of Wallace (1958c). The pore diameter in Eq. (1) was used because the complete pore diameter could not be measured. The glass beads had a simple pore structure with a uniform diameter larger than 130 μm. The sand had rather large pores with 20–130 μm diameter pores comprising 62.9% of the total number of pores. On the other hand, although andisol had a high abundance of pores, micro pores with diameter less than 20 μm constituted as much as 67.4% of total number of pores. The volume of pores over 20 μm in diameter was less in andisol than in glass beads or sand.

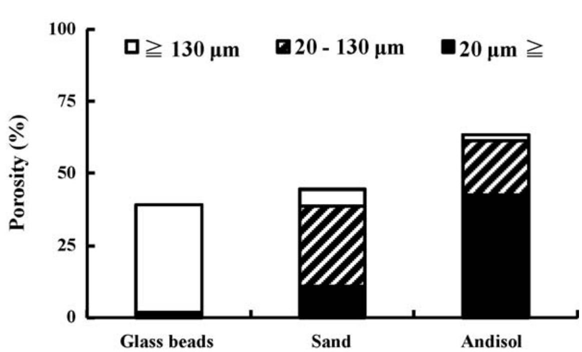

Fig. 1. Pore size distribution of glass beads, sand or andisol used to fill the experimental columns for assessing the movement of *Meloidogyne incognita* J2 nematodes. Sum of each pore size for each material shows the porosity (%) and the amount required for 1 pore volume per unit filling volume in this experiment.

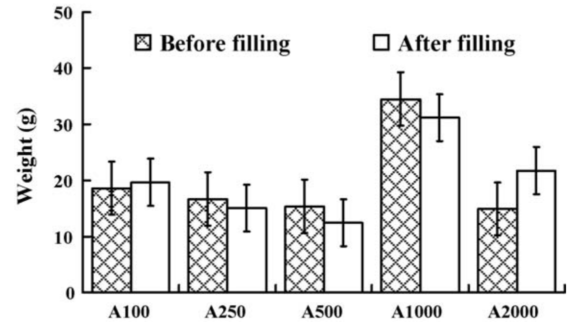

Fig. 2. The composition of aggregates in the andisol before and after filling the experimental columns used for assessing the movement of *Meloidogyne incognita* J2 nematodes. Aggregates were classified into five fractions according to size: A100, >100 μm; A250, 100–250 μm; A500, 250–500 μm; A1000, 500–1000 μm; A2000, 1000–2000 μm. Bars represent standard errors.

Fig. 2 shows the composition of aggregates in andisol. Aggregates in andisol were classified into five fractions according to size. The aggregate composition did not significantly change after filling the column (Scheffé's test, $p < 0.05$). The weight of the aggregates with diameter ranging from 500 μm to 1000 μm (A1000) was significantly greater than any other fraction after being filled into the experimental column (Scheffé's test, $p < 0.01$).

2.2. Concept of percolation

In this study, the volume of percolation is defined as "pore volume (PV, calculated by Eq. (2))". PV can be considered as a non-dimensional parameter to show the ratio of total volume of drainage from the column ($q_w tA$) to volume of the initial soil water in the column (θLA).

$$\mathrm{PV} = \frac{q_w tA}{LA} = \frac{ut}{L}, \tag{2}$$

where q_w is the flux, u the average pore water velocity, t the drainage time, and L is the length of the column. The value 'A' is the cross section of the column, but it can be cancelled since both denominator and numerator have the value A. When the soil is saturated with water, 1 PV equals the volume of initial soil water in the column. Thus PV is the water volume required to replace soil water from the inflow of the column to the outflow point.

2.3. Column experiment

The experimental device used is shown in Fig. 3. The column, made of acrylic resin, was a cylinder with inside diameter, cross section and length of 1.95 cm, 3 cm^2 and 11 cm, respectively. Glass beads, sand or andisol were used to fill the columns with bulk densities of 1.53 Mg m^{-3}, 1.62 Mg m^{-3} or 0.70 Mg m^{-3}, respectively. It was impossible to fill the three materials by the same density because of differences in soil particle density and grain diameter. After filling the column with each material, both ends of the column were capped with a metal mesh (mesh openings of 180 μm in diameter). This mesh allowed J2 to pass through because J2 width was 12.3–15.5 μm ($n = 30$, CV = 0.07) in our

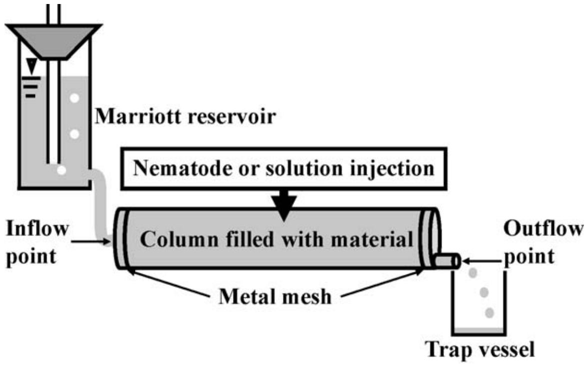

Fig. 3. Scheme of experimental device use for assessing the movement of *Meloidogyne incognita* J2 nematodes in different materials. The column was a cylinder made of acrylic resin and filled with glass beads, sand or andisol. Both ends of the column were capped with metal meshes. A Marriot reservoir to control water pressure was attached to the inflow point, and the 'trap' vessel was set at the outflow point to collect drainage water. Living J2, dead J2, or KBr solution was injected into the center of the column.

measurements. A Marriott reservoir was connected to the end at the inflow point and a metal pipe was connected at the outflow point leading into a 'trap' vessel.

The pressure head was positive between the Marriott reservoir and the column, and was zero between the inflow and outflow points of the column. Thus, the column was kept saturated with soil water throughout the experiment. It was impossible to unite flow velocity in the column because the porosity and the pore structure (presence of aggregates) were different. So, the soil water fluxes at the inflow and outflow point of the three materials were united. The soil water flux was adjusted to 1 drip/10 s (0.003 g/s, 36 cm h^{-1}) by changing the pressure head between the Marriott reservoir and the column. The experimental periods, flux, and flow velocity are described in Table 1. Under a constant flux, 1 ml of water suspension containing 500 $\pm$ 50 live J2, 500 $\pm$ 50 microwave-killed (600 W, 1 min) J2 or an equal volume of 5 mM KBr solution was injected into the center of the column. The shapes of microwave-killed J2s were almost straight. The shapes of a proportion of killed J2s were serpentine. KBr solution containing bromine ion (Br^-) was used as a tracer. In the andisol experiments KBr tracer was not employed because andisol was reported to absorb ions and may involve unexpected effects (Ishiguro, 2005).

All experiments were conducted at 20 °C constant temperature. Each experiment was repeated three times for all filling materials. The drainage water in the trap vessel was collected at 0.1 PV intervals until a total of 2.0 PV was collected. The J2 present in drainage water were counted under a stereomicroscope. The number of accumulated J2 thus obtained was divided by the initial number of J2 injected (C/C_0), to calculate a detection ratio. The J2 found at the outflow point were counted as "C". The J2 present on the meshes at the inflow and outflow points were also collected after finishing the experiment and counted. Movement of KBr was tested in a similar manner; Br^- concentration in drainage was quantified by ion chromatography, accumulated every 0.1 PV, and divided by the amount of injected KBr to obtain C/C_0 values.

Table 1
Experimental conditions in columns filled with glass beads, sand or andisol used to study nematode movement.

Type of filling	1 Pore volume (ml)	Experiment time (s)	Flux (cm h^{-1})	Flow velocity (cm h^{-1})
Glass beads	6.4	4200	36	92.3
Sand	7.3	4800	36	81.8
Andisol	10.4	6800	36	57.1

2.4. Statistical analyses

Detection ratios of living and dead J2 at the outflow point were adopted to compare the difference of materials and activity of J2 (Sokal and Rohlf, 1995). Statistical analyses used were two-way ANOVA followed by the Scheffe's test and Student's *t*-test to estimate the influence of the different materials and activity of J2. The mean of the dependent variables was arcsine transformed to obtain normality and homoscedasticity. These analyses were performed using JMP (Version 7.0.1, SAS Institute Inc., Cary, NC, USA). The Scheffe's test was also done among living J2, dead J2 and Br^- in each material.

3. Results

3.1. Movement of living and dead nematodes

Fig. 4 shows the movement of living and dead J2 nematodes in three different filling materials as fluctuation of C/C_0 values at various PV levels. Multiple comparisons were made of the number of J2 detected in each material. Table 2 shows the number of the nematodes present on the metal meshes at inflow and outflow points. We did not observe any J2 adhering to the columns made of acrylic resin.

According to ANOVA, the different filling materials, the activity of J2 and their interaction influenced the detection ratio of J2 at outflow point (Table 3). There was no difference between the glass

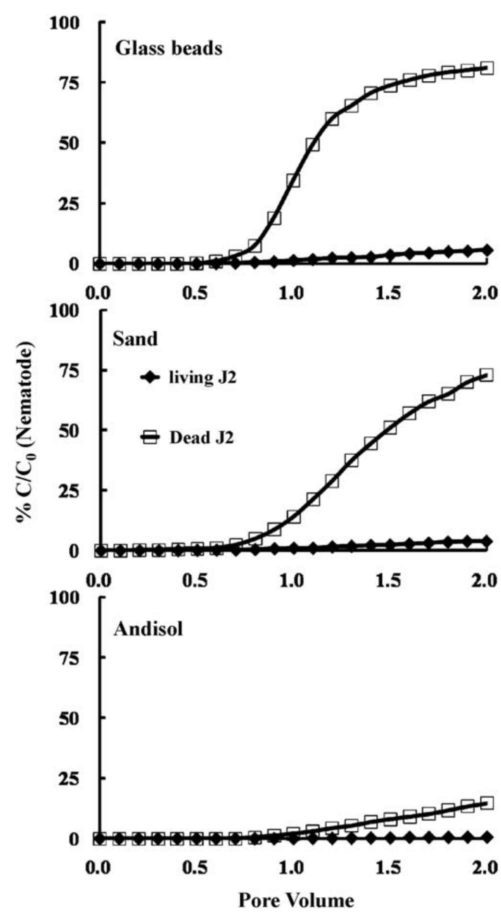

Fig. 4. Detection ratio of living (closed diamond) or dead (open square) *Meloidogyne incognita* J2 nematodes in columns filled with glass beads, sand or andisol. Each value is the mean of three replications. There was no difference between the glass beads and sand, but there was a significant difference between andisol and other materials (Scheffe's test, $p < 0.001$). There was a significant difference between living and dead nematodes in each material (Scheffe's test, $p < 0.001$).

Table 2
The numbers of *Meloidogyne incognita* J2 present on the metal meshes at inflow and outflow points of columns filled with different materials and inoculated with live or dead nematodes.

Type of filling	Living J2[a]		Dead J2[a]	
	Inflow point	Outflow point	Inflow point	Outflow point
Glass beads	0.0	8.7a	0.0	36.3a
Sand	0.0	7.7a	0.0	39.0a
Andisol	0.0	1.7b	0.0	10.7b

Different letters within living J2 or dead J2 indicate significant differences ($p < 0.05$).
[a] Values are means of three replications.

Table 3
ANOVA for the effect on the movement of live (active) or dead nematodes (*Meloidogyne incognita* J2) in columns with different filling materials.

Treatment factor	d.f.	SS	F	p
Material	2	921.790	23.575	<0.0001
Activity	1	3032.719	155.125	<0.0001
Material × activity	2	285.481	7.301	0.0008
Error	372	7272.671		

beads and sand but there was a significant difference between andisol and the other materials (Scheffe's test, $p < 0.001$). There was a significant difference between living and dead nematodes in each material (Student's t-test, $p < 0.001$).

In columns filled with glass beads, the numbers of living J2 that were present in the trap vessel showed a slight change with an increasing in PV to reach a number of 28 (7%) at 2.0 PV (Fig. 4), and the number of those present on the metal mesh at the outflow point averaged 8.7 (Table 2). On the other hand, when dead J2 were introduced, they started to appear in the drainage water from 0.6 PV and the number drastically increased from 0.9 to 1.2 PV (Fig. 4). At 2.0 PV a total of 405 J2 were present in drainage water with 36.3 J2 present on the mesh at the outflow point (Fig. 4 and Table 2), which represents almost 90% of the initial J2.

When living nematodes were introduced into columns filled with sand, 19 J2 were found in the drainage water after water flow of 2.0 PV and 7.7 J2 on the metal mesh at the outflow point (Fig. 4 and Table 2). As for dead J2, the number present in drainage water gradually increased at PV more than 0.7, and reached 364.3 (72.9%) at 2.0 PV with 39.0 J2 present on the metal mesh. Therefore over 80% of the initial dead J2 passed through the sand-filled column.

In andisol, the movement of both living and dead J2 was much lower than in the other two fillings (Fig. 4 and Table 2, Scheffe's test, $p < 0.01$). The sum of the living J2 in the drainage water and present on the metal mesh was 5 or less throughout the experiment. Also, of the initial dead J2 injected, 73.3 nematodes (14.7%) were present in drainage water and 10.7 (2.1%) were present on the metal mesh at the outflow point.

No J2 was found present on the metal mesh at the inflow point of any filling material (Table 2).

3.2. Solute transport

The results of solute transport experiments using columns filled with glass beads or sand are shown in Fig. 5. Andisol was not used in this experiment because of its ability to absorb ions and exchange cations.

The filling type did not affect the movement of Br^- in the columns by Student's t-test. The patterns of Br^- C/C_0 fluctuation were similar in columns filled with glass beads or sand. More than 90% of Br^- was collected between 0.7 PV and 1.2 PV irrespective of filling material.

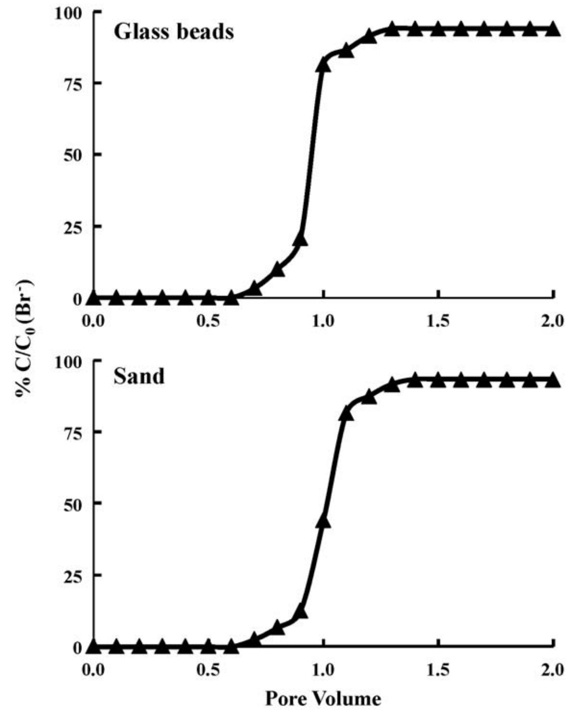

Fig. 5. Detection ratio of bromine ion (Br^-) which served as a passively transported solute in columns filled with glass beads or sand. Each value is the mean of three replications. There was no difference between the glass beads and sand (Student's t-test).

4. Discussion

Non-active ion (Br^-) moved according to convection dispersion theory both in glass beads and sand (Fig. 5), which is consistent with past reports (White et al., 1984; Khan and Jury, 1990; Fortin et al., 1997). Thus, the movement of passively transported ions under a soil water flow could be explained by a convection dispersion equation. However, it is likely not to have agreed with this equation if andisol was used, because andisol readily adsorbs the non-active ion, and therefore it would not be transported in the drainage water.

The living and dead J2 of *M. incognita* differed in movement in columns filled with all three types of materials, especially under the highest flow velocity (Fig. 4 and Table 2). More dead than living J2 passed through the column irrespective of the filling.

These results suggest that living J2 moved against flow direction or lodged on soil particle surfaces in columns with even very high flow velocity, although dense glass beads have many pores with diameters that are not optimal for J2 movement (Fig. 1; Wallace, 1958c). It was reported that *Escherichia coli* did not always follow percolation and was transported only through pores greater than 10–15 μm (White, 1985). The body width of J2 is 5–10 times greater than that of *E. coli*, and the nematodes could therefore be expected to move in water flows when the pore diameter exceeds 150 μm. In the present study, however, more than 90% of the living J2 remained inside the column without being transported to the outflow point by the water flow even in glass beads which comprised numerous pores of various sizes. Living J2s have irregular shapes that constantly change with nematode movement. This ability may allow J2 to assume a shape that can lodge in soil pores resisting the water flow. Most dead J2 have straight shapes that may pass through soil pores more easily. As for passively transported dead J2, the movement was not different for the column materials tested. The number of dead J2 in drainage water sharply increased in columns filled with glass beads and sand (Fig. 4). In contrast, in spite of such a difference in pore distribution, the movement of living J2 was similar between the two fillings. Therefore J2 may be able to resist movement even in macropores that are not considered to be optimal for nematode migration.

Flow velocity, which indicates water flow speed in columns filled with glass beads, sand or andisol, were 92.3 cm h^{-1}, 81.8 cm h^{-1} or 57.1 cm h^{-1}, corresponding to 22,152 mm d^{-1}, 19,632 mm d^{-1}, and 13,704 mm d^{-1}, respectively (Table 1). In our experiments living J2 were not transported into the drainage water even by such velocities, suggesting that they can withstand most natural soil water flows (several mm/h). With *Radopholus similis* under simulated rainfall, Chabrier et al. (2008) recently observed a similar phenomenon as demonstrated in our experiments, and they concluded that *R. similis* has developed behaviour to resist water flows. Although the nematode species may differ, our results supported their conclusion. Furthermore, our findings may suggest that other plant-parasitic nematodes have the ability of resisting or escaping movement in soil water flow, which may serve to increase their probability of infecting host plants.

In contrast, the dead nematodes transported into the drainage water were first found at 0.6 PV from the glass beads column and 0.7 PV from the sand column, inversely proportional to the flow velocity; and began to increase when the pore volume reached around 1.0 PV, in both tests (Fig. 4). Passive transport of ions in soil is generally discharged in 1.0 PV of drainage (White et al., 1984; Fortin et al., 1997), and, in the present study, bromine ion Br^{-} was discharged in around 1.0 PV of drainage in both glass bead- and sand-filled columns (Fig. 5). Migration of dead nematodes in single-grained material was similar to that of Br^{-} by the Scheffe's test, which may be calculated with the convection–dispersion equation of Bigger and Nielsen (1967). This convection–dispersion equation is widely used to predict solute transport in soils. The mechanical dispersion is described with an equation mathematically identical to Fick's law for molecular diffusion. Physically unrealistic backward solute mixing may occur for the convection–dispersion equation when a high concentration gradient exists in a soil. The non-active ion moves in the direction of the flow in case of convection. Moreover, the movement by convection differs depending on the kind of ion, and it is corrected by the coefficient. However, dead nematodes do not disperse by a concentration gradient and they cannot move in any pore diameter, as they are organisms. So, dead nematodes may follow a coefficient of convection of their original, but an optimized coefficient could not be found by inverse analysis.

The pattern of J2 transport in andisol was distinct from the other two fillings and most of the nematodes injected remained in the column even in the experiments using dead nematodes (Fig. 4). This may be due to the aggregate structure, complicated pore size distribution, and the resulting tortuous water flow in andisol (Figs. 2 and 3). Andisol has a huge numbers of pores with various sizes and complicated distribution. The diameters of the aggregates were also divergent within 2000 μm. Assuming J2 in general cannot pass through pores less than 20 μm (Wallace, 1958c), it is likely that over 60% of the andisol bulk does not permit J2 transport. This is consistent with our result that the nematodes hardly were present in drainage water of andisol-filled columns.

Our study presented here demonstrated that J2 can remain in distinctive types of soil and resist a water flow. Future study needs to clarify how they resist water flow and how they sense the surrounding soil environment. This may provide us with more accurate and efficient measures to control *M. incognita*.

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